



ρ_{DCCA} applied between air temperature and relative humidity: An hour/hour view

G.F. Zebende^{a,d,*}, A.A. Brito^{b,c}, A.M. Silva Filho^d, A.P. Castro^e

^a Department of Physics, State University of Feira de Santana, Bahia, Brazil

^b Federal Institute, Paulo Afonso, Bahia, Brazil

^c SENAI CIMATEC, Salvador, Bahia, Brazil

^d Earth Sciences and Environment Modeling Program, State University of Feira de Santana, Bahia, Brazil

^e Jorge Amado University Center, Salvador, Bahia, Brazil

HIGHLIGHTS

- We quantify cross-correlations between air temperature and relative humidity.
- Our data were available hourly by the Brazilian Institute of Meteorology (INMET).
- We present a new vision about these variables at every scales with a detrended cross-correlation map.

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ABSTRACT

In this paper we propose to study the cross-correlation between the air temperature and the air relative humidity using the DCCA cross-correlation coefficient ρ_{DCCA} . We choose four cities (meteorological stations) located in the State of Bahia (Brazil), as our case study. Our data were available hourly from the Brazilian Institute of Meteorology. Accordingly, in addition to showing that the variables have a negative cross-correlation, we present a new vision about these variables, producing the detrended cross-correlation contour map. As a result, we can see (or not) islands of cross-correlations, that depend on the local time and also the time scale involved.

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1. Introduction

Global warming can be caused by human activities (anthropogenic causes), as well as, by natural phenomena. The climate components are never in equilibrium and are constantly varying, like a complex system, see [1–4]. Thereby, any change in the components may result in a considerable variation in the climate [5]. From the perspective of climatology, mathematical models are important, because they can be applied to study the dynamics of the weather, projections of the future climate, and changes in the air temperature, among others. Meteorological data accumulated over many years, such as records of rainfall, temperature, and atmospheric composition can be analyzed by modern statistical techniques and mathematical models [6–10]. Thus, to study these complex systems we must properly define the fundamental variables, such as: air temperature, pressure, wind direction and speed, relative humidity, and many others, see chapter five of [11]. Most of these models use systems of differential equations based on the laws of physics, fluid motion, and chemistry. But, [12] recently proposed studying the cross-correlation between two time series by applying the DCCA method, designed to investigate

* Corresponding author at: Department of Physics, State University of Feira de Santana, Bahia, Brazil.

E-mail address: gzebede@pq.cnpq.br (G.F. Zebende).

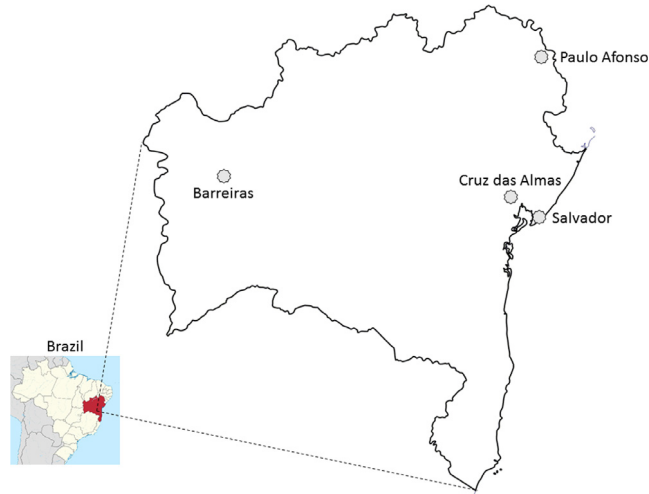


Fig. 1. State of Bahia (BR) and four climatological stations located in cities of: Barreiras, Cruz das Almas, Paulo Afonso, and Salvador.

Table 1

Geographical information for the stations.

Station	Lat.(S)	Long.(E)	Elev.(m)
Barreiras	12°07'29.0"	45°01'37.4"	474
Cruz das Almas	12°40'31.5"	39°05'22.5"	220
Paulo Afonso	09°22'41.2"	38°13'36.4"	255
Salvador	13°00'19.9"	38°30'20.7"	48

Table 2

Data information.

Station	Start	End	points
Barreiras	12/22/2001	09/30/2010	2047
Cruz das Almas	01/26/2003	09/30/2010	1919
Paulo Afonso	06/18/2003	09/30/2010	2065
Salvador	10/07/2000	09/30/2010	2616

power-law cross-correlations between different simultaneously recorded time series in the presence of non-stationarity. However, the DCCA method does not quantify the level of cross-correlation. In order to solve this problem [13] defined the DCCA cross-correlation coefficient, ρ_{DCCA} , which allows the analysis of non-stationary time series, as in climatological variables. To show the efficiency of the DCCA cross-correlation coefficient, we analyze the relationship between the air temperature and the relative humidity. Our database comes from four stations located in Bahia State (BR), with data measured hourly by the Brazilian Institute of Meteorology (INMET).

With an area of 564,733.177 km², Bahia is one of the 27 federated Brazilian States, bordering eight other States. On the east it is bathed by the Atlantic Ocean, having the most extensive coast in Brazil. Below, in Fig. 1, we present the location of the meteorological stations, and their description in Table 1.

The importance of these stations (cities) comes from the fact that they are the ones that have the best databases, and because they have the following characteristics (see [14] and Table 2):

- **Barreiras:** This is the most populous city and the most important agricultural center in the western region of Bahia. The city is pierced by the Rio Grande river, the main affluent on the left of the São Francisco river (one of the most important in Brazil and South America). Barreiras stands out in national (international) agribusiness as a great producer of cotton and soybeans.
- **Cruz das Almas:** This is considered an important center in the *Recôncavo*, a geographical region located around the *Todos os Santos* bay, covering not only the coast but also the entire region of the interior surrounding the bay. The city has some agricultural research centers, such as Brazilian Company of Agricultural Research (EMBRAPA), and its industry is known for well made footwear and textiles.
- **Paulo Afonso:** This is the city where the São Francisco Hydroelectric Company (CHESF) plant complex is located. It has four large plants, PA I, PA II, PA III, and PA IV. This makes Paulo Afonso known as the “energy capital”. The city has a planned structure, the center is inside an artificial island that was built with the implantation of the channel of the

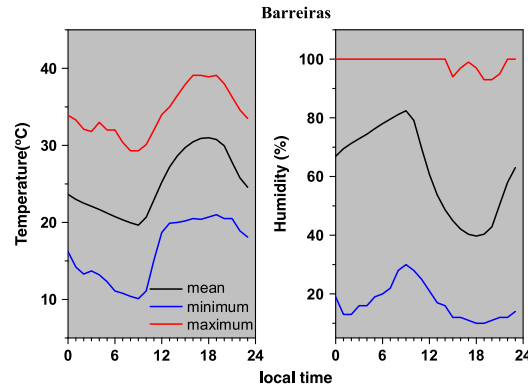


Fig. 2. (color on-line) Descriptive statistic with mean, minimum, and maximum values of the air temperature and the relative humidity for Barreiras.

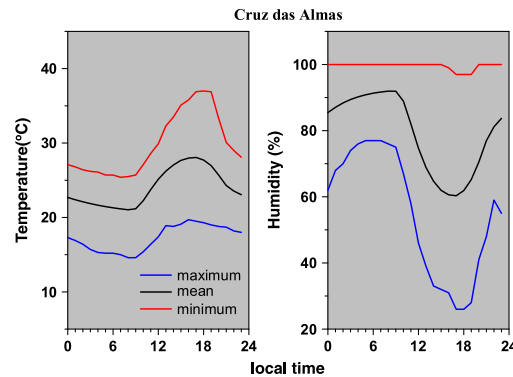


Fig. 3. (color on-line) Descriptive statistic with mean, minimum, and maximum values of the air temperature and the relative humidity for Cruz das Almas.

PA IV plant. The main characteristic of the hydroelectric plant is to have been the first underground plant installed in Brazil, because its turbines are more than 80 m below the level of the São Francisco river.

- **Salvador:** This is the capital and economic center of Bahia, and is notable for its food, music and architecture, with strong African and Portuguese influences. With more than 2.9 million people, it is the most populous city in the Northeast and the third most populous in Brazil.

Thus, taking into account the number of meteorological variables and the wide possibility of its applications, in this paper we restrict ourselves to studying the cross-correlation between the air temperature and the air relative humidity [15–20]. We adopt the DCCA cross-correlation coefficient ρ_{DCCA} [13], which is defined in terms of the DFA [21] and the DCCA method [12], briefly described in Section 3.

2. Data and descriptive statistics

Our data was made available by INMET [22], and measured hourly over a period that ranges generally from 2001 to 2010, see Table 2. Initially we present the descriptive statistics of the data with: mean, minimum, and maximum values of the air temperature and relative humidity, presented in Figs. 2–5 as a function of the local time. We can see that there is an inverse trend between these variables, an increase (decrease) in the air temperature leads to a decrease (increase) in the relative humidity.

Specifically for Barreiras, Fig. 2, we note a smaller air temperature and a greater relative humidity at 9:00. But, at approximately 17:00, the air temperature is maximal and the relative humidity is minimal. Cruz das Almas and Paulo Afonso, Figs. 3 and 4, are similar to Barreiras. However, there is also present an inverse form for Salvador, the air temperature and the relative humidity have a lower variability, see Fig. 5. These variables can be seen also a functions of time with a contour plot, see Figs. 6–9.

In general, we can see that the air temperature ranges from 15 °C to 40 °C and the relative humidity from 10% to 100%. From these figures, we can conclude that there is apparently an inverse relation between the air temperature and the relative humidity, and that Salvador presents less variability, if compared to the other three cities. However, if we want to measure

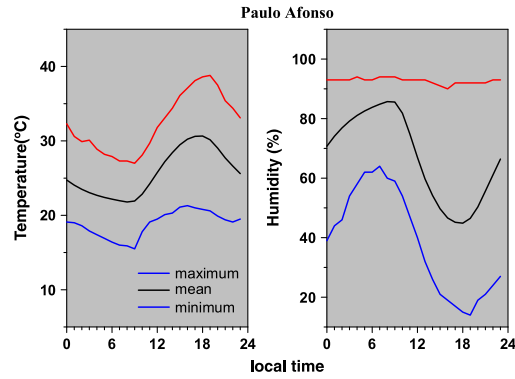


Fig. 4. (color on-line) Descriptive statistic with mean, minimum, and maximum values of the air temperature and the relative humidity for Paulo Afonso.

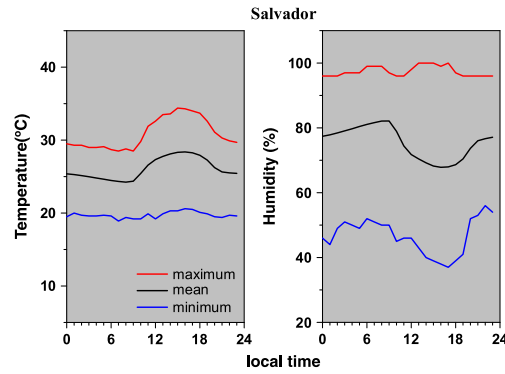


Fig. 5. (color on-line) Descriptive statistic with mean, minimum, and maximum values of the air temperature and the relative humidity for Salvador.

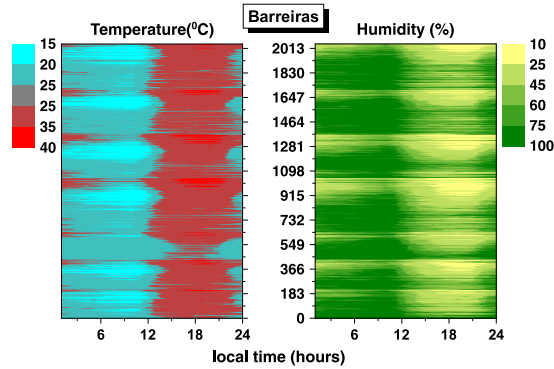


Fig. 6. (color on-line) Contour plot of the air temperature and relative humidity for Barreiras. The vertical axis shows the time (in days) and horizontal axis shows the local time.

the cross-relation between these two variables and verify whether or not there is an inverse relation between them, we should apply a statistical test that is well-adjusted for non-stationary time series analysis. Thus, the proposal in this paper is to apply the detrended cross-correlation coefficient, ρ_{DCCA} , which measures the cross-correlation between two non-stationary time series, briefly described below.

3. Discussion and methods

In time series analysis there are some well-known paths to follow, [23,24]. Thus, if the time series exhibit complex behavior such as self-affinity, we can apply new strategies for its analysis [25–30]. From this point of view, one of the most frequently cited methods for analyzing the time series of complex problems is detrended fluctuation analysis (DFA) [21]. This

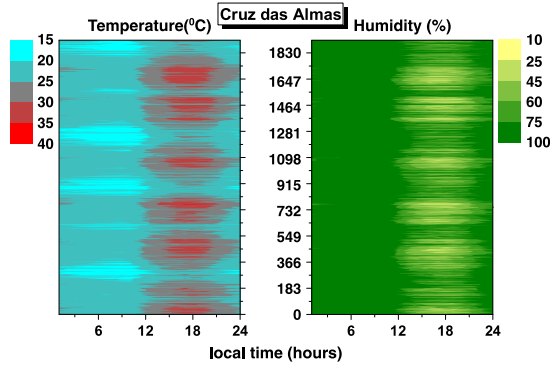


Fig. 7. (color on-line) Contour plot of the air temperature and relative humidity for Cruz das Almas. The vertical axis shows the time (in days) and the horizontal axis show the local time.

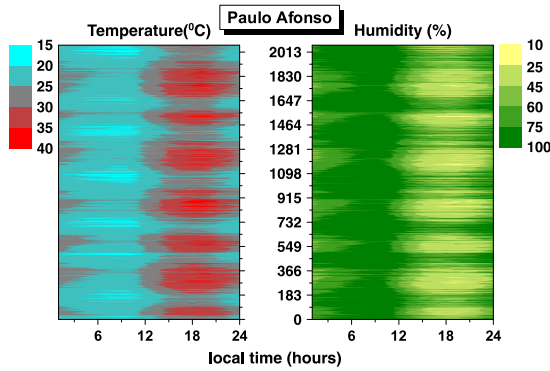


Fig. 8. (color on-line) Contour plot of the air temperature and the relative humidity for Paulo Afonso. The vertical axis shows the time (in days) and the horizontal axis show the local time.

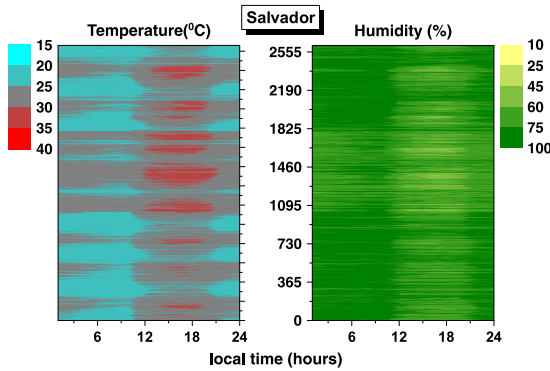


Fig. 9. (color on-line) Contour plot of the air temperature and the relative humidity for Salvador. The vertical axis shows the time (in days) and the horizontal axis show the local time.

method provides a relation between $F_{DFA}(n)$ (root mean square fluctuation) and the scale n . The DFA method has been very efficient at detecting long-range auto-correlations embedded in a patch landscape and also avoiding the spurious detection of apparent long-range auto-correlations. This can be proved by a great number of applications and citations [31–34]. However, if we have two time series, $\{y_i\}$ and $\{y'_i\}$, the analysis of cross-correlation can be applied, as in [35–37]. Here, we will adopt the detrended cross-correlation coefficient of [13], based on the Detrended Cross-Correlation Analysis method (DCCA) [12].

The DCCA method is a generalization of the DFA method and is based on detrended covariance. This method is designed to investigate power-law cross-correlations between different simultaneously recorded time series in the presence of nonstationarity. Therefore, for two time series of equal length N , we compute two integrated signals $R_k \equiv \sum_{i=1}^k y_i$ and

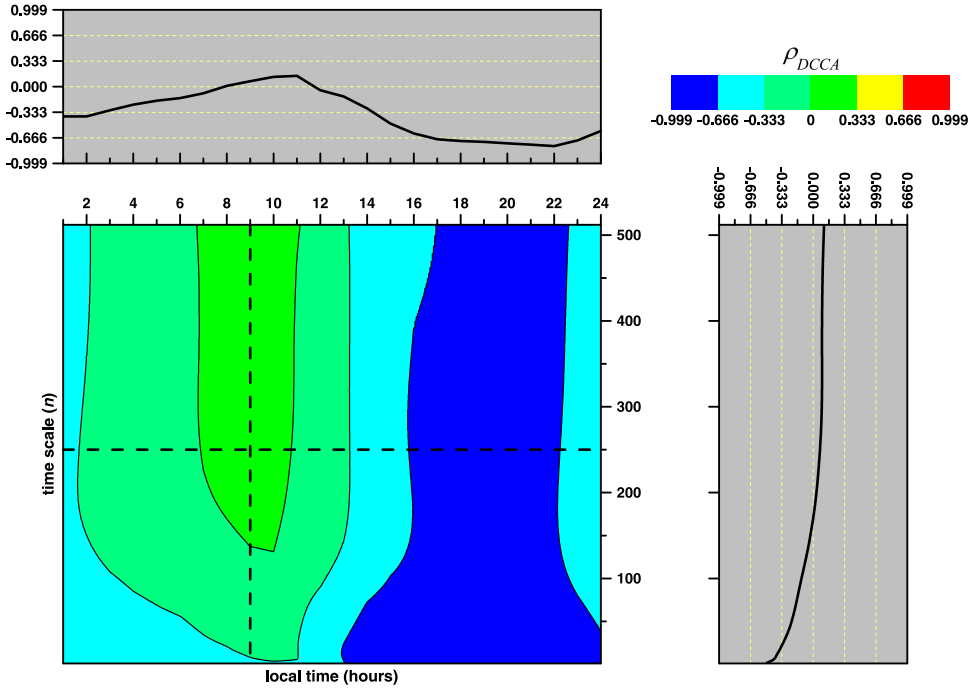


Fig. 10. (color on-line) Contour plot of ρ_{DCCA} for Barreiras. Horizontal line represent the local time at 9:00 and vertical line represent $n = 250$ (time scale).

$R'_k \equiv \sum_{i=1}^k y'_i$, where $k = 1, \dots, N$. In the next step we divide the entire time series into $(N - n)$ overlapping boxes, each containing $(n + 1)$ values. For both time series, in each box that starts at i and ends at $(i + n)$, we define the local trend, $\tilde{R}_{k,i}$ and $\tilde{R}'_{k,i}$ ($i \leq k \leq i + n$), to be the ordinate of a linear least-squares fit. We define the detrended walk as the difference between the original walk and the local trend. Next we calculate the covariance of the residuals in each box $f_{DCCA}^2(n, i) \equiv 1/(n + 1) \sum_{k=i}^{i+n} (R_k - \tilde{R}_{k,i})(R'_k - \tilde{R}'_{k,i})$. Finally, we calculate the detrended covariance function by summing over all overlapping $N - n$ boxes of size n :

$$F_{DCCA}^2(n) \equiv (N - n)^{-1} \sum_{i=1}^{N-n} f_{DCCA}^2(n, i). \quad (1)$$

When only one random walk is analyzed, ($R_k = R'_k$), the detrended covariance $F_{DCCA}^2(n)$ reduces to the detrended variance $F_{DFA}^2(n)$, used in the DFA method. If there is self-affinity, then $F_{DCCA}^2(n) \sim n^{2\lambda}$. DCCA has been applied in many situations [38–42]. The λ exponent quantifies the long-range power-law cross-correlations and also identifies seasonality, [42], but λ does not quantify the level of cross-correlation.

To quantify the level of cross-correlation, we can apply the DCCA cross-correlation coefficient, [13], defined as the ratio between the detrended covariance function F_{DCCA}^2 and the detrended variance function F_{DFA} , i.e.,

$$\rho_{DCCA}(n) \equiv \frac{F_{DCCA}^2(n)}{F_{DFA(y_i)}(n) F_{DFA(y'_i)}(n)}. \quad (2)$$

Eq. (2) leads us to a new scale of cross-correlation in non-stationary time series. The value of ρ_{DCCA} ranges between

$$-1 \leq \rho_{DCCA} \leq 1.$$

In this case, $\rho_{DCCA} = 0$ means there is no cross-correlation, and it splits the level of cross-correlation between positive and the negative case (see Table 1 in [13]). The detrended cross-correlation exponent, ρ_{DCCA} , has been tested on selected time series, simulated and real cases, and has proved to be quite robust. See the link below for more details:

4. Results

In Section 2, we can see directly that the air temperature and the relative humidity are anti cross-correlated, but how much? The answer to this question will be given by calculating ρ_{DCCA} for our variables. Here we will present the final results in a contour map (a new idea), where the x-axis represents the local time and the y-axis the time in days, see Figs. 10–13.

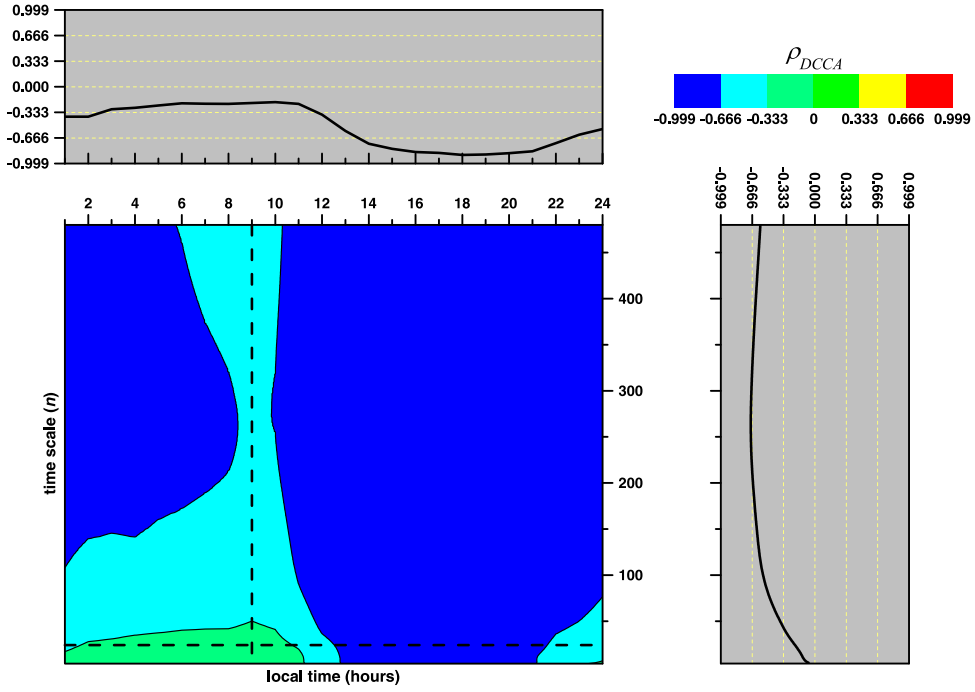


Fig. 11. (color on-line) Contour plot of ρ_{DCCA} for Cruz das Almas. Horizontal line represent the local time at 9:00 and vertical line represent $n = 24$ (time scale).

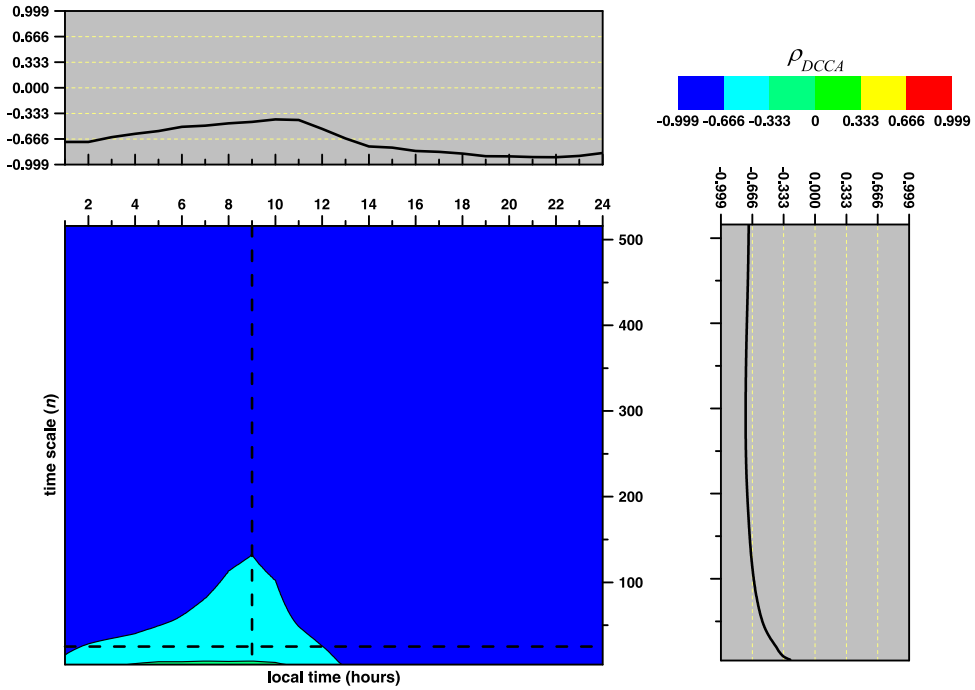


Fig. 12. (color on-line) Contour plot of ρ_{DCCA} for Paulo Afonso. Horizontal line represent the local time at 9:00 and vertical line represent $n = 25$ (time scale).

Thus, based on Table 3 we can define, by colors, where there is a weak, medium or strong cross-correlation (negative or positive).

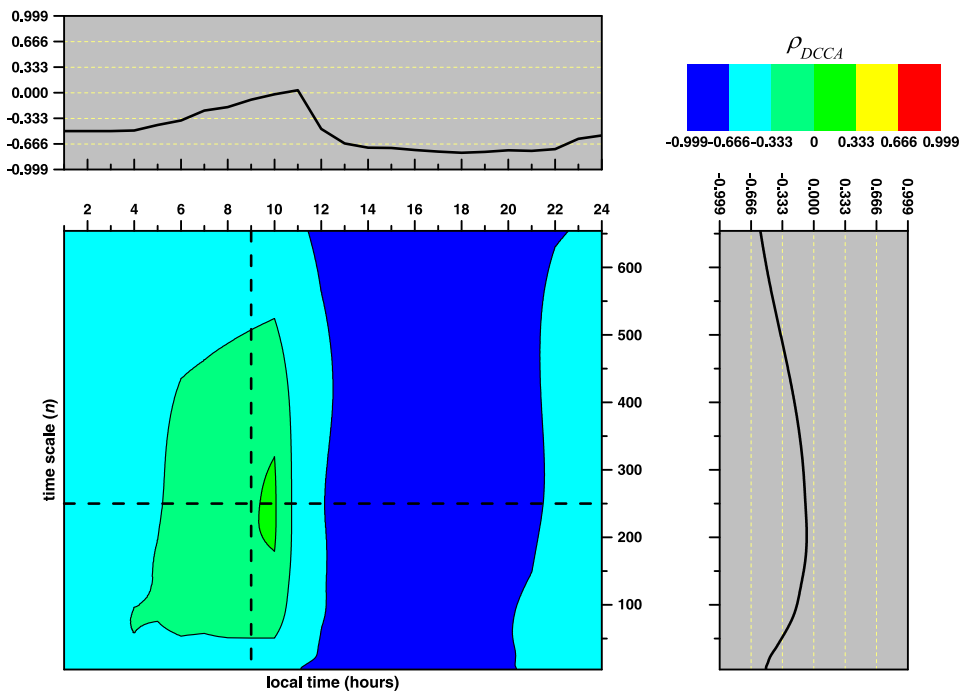


Fig. 13. (color on-line) Contour plot of ρ_{DCCA} for Salvador. Horizontal line represent the local time at 9:00 and vertical line represent $n = 250$ (time scale).

Table 3

Detrended cross-correlation coefficient intervals.

Source: [http://scholar.google.com.br/scholar?oi=bibs\(&\)hl=pt-BR{&}cites=6351233990136499122](http://scholar.google.com.br/scholar?oi=bibs(&)hl=pt-BR{&}cites=6351233990136499122).

Condition	ρ_{DCCA}
Weak	$\pm 0.000 \mapsto \pm 0.333$
Medium	$\pm 0.333 \mapsto \pm 0.666$
Strong	$\pm 0.666 \mapsto \pm 0.999$

According to the six levels defined (three positive and three negative) in this paper, we can associate one color to a range of values of ρ_{DCCA} . In our case, the value of the detrended cross-correlation coefficient goes from the most negative value (blue) to the most positive (red). In this way, we have a complete climatological map using the level of cross-correlation between these two variables, the air temperature and the relative humidity.

Looking at the climatological map of ρ_{DCCA} and depending on the color position, we can initially clearly see that there are “islands” of cross-correlations, a fact not yet seen in the previous literature. For example, Fig. 10 shows these islands of cross-correlation mainly in the vertical direction (temporal scale). In the horizontal direction (local time), we can see transitions of anti cross-correlation (negative case) from weak (small portion), medium (during the morning) to strong (afternoon and evening). Furthermore, as an example, we present ρ_{DCCA} as a function of a specific scale: horizontal (local time) and vertical (time scale) dashed lines. This highlights the cross-correlation transition between the levels.

For Cruz das Almas, Fig. 11, there is a larger island of strong anti cross-correlation (blue) level, and a small portion of a weak anti cross-correlation level, located at $n \approx 20$ and between 0 h and 10 h. For Paulo Afonso station, Fig. 12, what is present is almost entirely an anti cross-correlation strong (blue), except for small temporal scales at the beginning of the day where medium anti-cross-correlation show up. Salvador, Fig. 13, has some similarities with Barreiras, regarding vertical islands. The period between 12 h and 22 h shows a strong anti cross-correlation. For this station, near the coast, a small island appears around 10 h and $n \approx 250$, where the value of ρ_{DCCA} is slightly positive and weak.

5. Conclusions

In this paper we can see the temporal evolution of the air temperature and the relative humidity measured hourly for four automatic meteorological stations, located in Bahia State (Brazil). Comparing their mean values with approximately 10 years of measurements, we can see that there is an inverse relation between them. However, to quantify how the variables are anti cross-correlated, we have applied the detrended cross-correlation coefficient, ρ_{DCCA} , designed to measure the cross-correlation between two non-stationary time series. But the interesting and new fact is that we were able to see the value

of ρ_{DCCA} in a climatological map. From this map and taking into account some ranges of cross-correlation range, we can see directly in which regions there was strong, medium, and weak cross-correlation (negative and positive).

Specifically, for Barreiras, a strong anti cross-correlation can be seen between 16 h and 22 h. But, for other local times, this behavior tends to a smaller values, e.g., for mornings. If we compare Cruz das Almas with Barreiras, there is a larger portion of anti cross-correlation levels. Paulo Afonso station was the one that presented the highest anti cross-correlation (strong) levels for all time scales. Salvador, the station with the lowest variability in the air temperature and relative humidity, has a behavior similar to that of Barreiras in this sense of ρ_{DCCA} .

Finally, in this paper we could see the new climatological map based on the detrended cross-correlation coefficient between the air temperature and the relative humidity, two important climatological variables. This map can give us a complete view of these variables at all temporal scales.

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